

Evidence-First Astronomical Anomaly Investigation

Rahman Darji

September 2026

Abstract

Large astronomical surveys provide repeated measurements of enormous numbers of sources, making automated methods necessary for identifying objects whose observed behavior differs from the dominant population. However, an unusual measurement is not necessarily an astrophysical discovery. Apparent anomalies can arise from measurement uncertainty, temporal sampling, source confusion, extraction effects, image artifacts, calibration behavior, or other observational systematics. A reliable anomaly-investigation workflow therefore needs to distinguish statistical unusualness from evidence that the unusual behavior is observationally credible, from the risk that it is produced by known systematics, and from any subsequent astrophysical interpretation.

This work presents an evidence-first framework for investigating unusual astronomical sources and demonstrates the framework on single-exposure NEOWISE W1 and W2 data from a small development field. The workflow treats a source, rather than an individual detection, as the unit of analysis and progressively combines temporal behavior, photometric consistency, image-level evidence, observational and systematic checks, external catalogue information, and comparison with known source populations. Statistical screening is used to reduce the search space, while later stages are explicitly prevented from being interpreted as automatic confirmation. Novelty, evidence quality, and systematic risk are maintained as distinct dimensions rather than collapsed into a single discovery score.

The investigation began with 8,652 retrieved detections. After quality and usability filtering and removal of the originally investigated source, 7,275 detections remained in the working sample, forming 3,371 source-like groups. A minimum-observation requirement produced 175 source histories suitable for the source-level analysis. A corrected internal validator identified six positive groups among 82 groups for which the validator was evaluable. This rate is an internal trigger rate within the constructed evaluation population and is not interpreted as a false-positive rate, prevalence, sensitivity, or specificity. Subsequent unsupervised analyses showed that statistical novelty could be strongly affected by brightness and signal-to-noise-related features, motivating feature-family and brightness-controlled diagnostics rather than direct interpretation of an anomaly score as astrophysical novelty.

Two sources, designated 1978 and 1974 within the internal source-group representation, survived as the highest-priority investigation cases. Source 1978 retained elevated W1 reduced chi-square under leave-one-out testing, was unusual relative to the matched control comparison, and showed source-like image-level signal. At the same time, its W2 behavior was less supportive, nearby sources remained a plausible source of contamination, and external identity checks were

incomplete. Source 1974 showed a similar but somewhat weaker pattern. Neither source is presented as a confirmed variable object, new class, or new astrophysical phenomenon. The principal result of the study is methodological: the investigation demonstrates how automated unusualness detection can be embedded within a staged evidence process that narrows a large search space while preserving unresolved alternatives and making the evidential status of each conclusion explicit.

1 Introduction

Modern astronomical surveys repeatedly observe enormous numbers of sources across wide areas of the sky. The resulting datasets contain information about variability, transient behavior, source populations, and changes in measured brightness that would be difficult to inspect manually at scale. Automated anomaly detection and machine-learning methods therefore provide an attractive way to reduce a large observational dataset to a smaller set of sources that warrant closer investigation. The central difficulty is that an observational anomaly and an astrophysical anomaly are not equivalent. A source may appear unusual because of measurement uncertainty, sparse or uneven temporal sampling, differences in photometric extraction, nearby-source contamination, image artifacts, calibration behavior, detector effects, or other properties of the observing system. Conversely, a genuinely interesting source may not be unusual under a single statistical representation. A system designed only to maximize an anomaly score can therefore produce a ranked list without establishing why the objects are unusual or whether the unusualness survives observational scrutiny.

This distinction motivates an evidence-first approach. In such an approach, statistical methods are used primarily to identify observations or source histories that deserve attention. Subsequent stages ask progressively different questions: Is the signal internally consistent? Is the source spatially present in the image? Does the behavior persist when individual observations are removed? Is the behavior reproduced in other bands? Is there evidence for a nearby contaminating source? Is there an obvious image artifact or observational pattern? Does an external catalogue provide an identity or known classification? Does the source resemble a known population? Which explanations remain unresolved?

The objective is consequently not to define a single number that represents how “interesting” a source is. Instead, the objective is to construct an auditable evidence profile in which different kinds of information retain their distinct roles. Statistical unusualness can be high while observational evidence remains weak. A source can have good data quality while still having a plausible systematic explanation. External information can be unavailable without implying that a source is novel. These distinctions are important because collapsing them into a single score can make uncertainty difficult to recover after the fact.

The scientific question addressed here is therefore:

Can an evidence-first, multi-stage investigation pipeline identify and prioritize unusual astronomical sources in NEOWISE single-exposure data while explicitly separating sta-

tistical unusualness from observational evidence, systematic risk, and astrophysical interpretation?

The study is deliberately framed as a methodological demonstration rather than a claim of a new astronomical discovery. The implementation is exercised on a small development field within the NEOWISE single-exposure data. The purpose of this setting is to expose the complete investigation chain, including candidate rejection, statistical diagnostics, image inspection, systematic checks, external-catalogue queries, and uncertainty handling. Because the analysis uses one survey and one small field, the results should not be interpreted as demonstrating equivalent performance across other surveys, wavelengths, source populations, or observing systems.

The main contribution is the organization of anomaly investigation into an explicit evidence chain:

unusual → data-quality checked → internally robust → observationally inspected → systematic alternatives tested → external information checked → candidate for further investigation.

This sequence is intentionally different from a discovery claim. A source can reach the final stage while still having unresolved explanations. In particular, the framework treats “unexplained” as a statement about the limits of the checks performed, not as evidence for an exotic physical interpretation.

The success criterion of the pipeline is consequently not the discovery of a new object. It is the ability to narrow a large search space to a substantially smaller investigation set while preserving provenance, uncertainty, alternative explanations, and the distinction between automated screening and scientific confirmation.

The remainder of the paper describes the investigation setting, the evidence-first framework, the statistical and machine-learning analyses, the observational vetting process, and the resulting candidate evidence profiles. The final sections discuss what the demonstration supports, what it does not establish, and which additional observations or independent datasets would be required before stronger astrophysical conclusions could be justified.

2 Related Work and Positioning

Automated discovery and classification are already established components of modern astronomical data analysis. Large surveys routinely use statistical selection, machine-learning methods, variability metrics, catalogue cross-matching, image-level information, and human inspection to identify objects requiring additional attention. The present work should therefore not be interpreted as introducing the general idea of automated anomaly detection or multi-stage astronomical vetting. A particularly relevant example is the VarWISE effort, which uses the long temporal baseline of NEOWISE single-exposure measurements to identify and classify infrared-variable sources. Such

work demonstrates the value of repeated W1 and W2 observations and machine-learning-based population analysis for extracting variability information from the same broad observational ecosystem used here.

The methodological emphasis in this study is different. Rather than treating classification or anomaly ranking as the endpoint, the framework separates several questions that are often conflated during exploratory investigations. First, is a source statistically unusual within the analyzed population? Second, does that unusualness remain after accounting for measurement uncertainty and sensitivity to individual observations? Third, is there observational evidence that the measured source is genuinely present in the images? Fourth, are known or plausible systematic explanations supported by the available data? Fifth, does external catalogue information identify the source or place it within a known population?

This distinction is related to work on active anomaly detection and human-in-the-loop astronomical analysis, where automated systems can reduce a large search space but still require domain-aware inspection. The present framework similarly treats machine learning as a prioritization mechanism rather than as an autonomous discovery or confirmation mechanism.

The contribution should therefore be understood as an evidence organization and investigation framework demonstrated on NEOWISE data. The analysis does not establish that the same implementation has equal effectiveness on optical time-domain surveys, radio surveys, X-ray observations, or other astronomical datasets. Such transferability would require independent evaluation using different instruments, populations, cadence structures, noise properties, and systematic-error regimes.

The distinction is also important for the interpretation of negative results. A candidate that fails an observational check is not merely an unsuccessful anomaly score; it provides information about which apparent anomalies can be explained by known observational effects. Conversely, a candidate that survives the available checks is not automatically a discovery. It is a source for which the tested explanations have not provided a satisfactory account at the current stage of investigation.

Accordingly, the framework is designed around progressive evidential narrowing rather than maximizing the number of apparently novel sources. This makes candidate rejection an informative outcome rather than a failure of the pipeline.

3 Data and Investigation Setting

3.1 NEOWISE Single-Exposure Data

The investigation uses single-exposure measurements from the NEOWISE survey. The analysis focuses on the two infrared bands W1 and W2, centered at approximately 3.4 and 4.6 μm , respectively. Repeated observations provide a temporal record for individual source positions and allow source-level quantities such as scatter, temporal range, and uncertainty-normalized deviations to be calculated.

The development field was restricted to approximately

$$179.9^\circ \leq \text{RA} \leq 180.1^\circ$$

and

$$-0.1^\circ \leq \text{Dec} \leq +0.1^\circ.$$

This deliberately small field was used to make the full investigation chain computationally manageable while still retaining repeated single-exposure observations and sufficient local structure for control and systematic analyses.

The working data include source coordinates, W1 and W2 magnitudes and their uncertainties, observation times, and frame-quality information. The principal detection-level quantities used in the analysis include

$$\text{RA}, \quad \text{Dec}, \quad W1, \quad \sigma_{W1}, \quad W2, \quad \sigma_{W2}, \quad \text{MJD}, \quad \text{qual_frame}.$$

The single-exposure nature of the data is important to the interpretation of the analysis. Individual measurements are affected by the observing conditions and measurement process associated with particular frames. Consequently, a source-level anomaly can reflect either intrinsic source behavior or a pattern introduced by the observation and extraction process. The evidence-first design explicitly retains this distinction throughout the investigation.

3.2 Quality Filtering and Source Construction

The initial retrieval contained 8,652 detections. Quality and usability filtering reduced this to 7,308 usable W1 measurements. After excluding the source associated with the original targeted anomaly investigation, 7,275 detections formed the working sample used for the broader source-level analysis. The surviving detections were grouped into source-like histories using their sky positions. This produced 3,371 source-like groups. Because very short histories provide limited information about temporal behavior, the main source-level analysis required at least 10 observations. Under this criterion, 175 source groups were available for the subsequent investigation.

The filtering and grouping process is important because the unit of analysis is a source history rather than an individual detection. A single unusually measured exposure can be informative, but it does not by itself establish that a source behaves unusually. Source-level aggregation allows the investigation to examine whether apparent deviations are distributed across a history, concentrated in a small number of observations, or associated with particular observing circumstances.

The observation count threshold was an operational screening criterion rather than a statistically optimized detection threshold. Later sensitivity checks explicitly examined how changing the minimum-observation threshold affected the validator results. This distinction prevents the operational threshold from being interpreted as an independently established boundary between real and spurious astronomical variability.

3.3 Source-Level Measurements

For each source group, basic temporal and photometric measurements were calculated from the available detections. These included the number of observations, temporal span, weighted mean magnitude, standard deviation, median, range, robust scatter estimates, median W1 signal-to-noise ratio, and reduced chi-square statistics.

For a measured quantity x_i with uncertainty σ_i , the uncertainty-normalized residual was represented as

$$z_i = \frac{x_i - \mu}{\sigma_i},$$

where μ is the reference level used for the source.

A reduced chi-square statistic was used as an uncertainty-aware measure of scatter:

$$\chi^2_\nu = \frac{1}{N-1} \sum_{i=1}^N \frac{(x_i - \mu)^2}{\sigma_i^2}.$$

These quantities are useful for identifying source histories whose observed scatter is large relative to their reported measurement uncertainties. They are not, by themselves, interpreted as proof of intrinsic astrophysical variability. In particular, a high reduced chi-square can arise when uncertainties do not fully describe all sources of observational variation, when measurements are correlated, or when the adopted reference model is inadequate.

The analysis therefore uses these statistics as screening and diagnostic quantities within a larger evidence chain. Later tests examine the sensitivity of elevated chi-square values to individual observations, compare sources with matched controls, inspect W1 and W2 behavior separately, and investigate observational explanations.

The source representation also retains explicit information about data availability. A missing image-level measurement, failed external catalogue query, or unavailable known-class comparison is not converted into a numerical value representing normality. Such cases are recorded as unavailable or UNKNOWN and are interpreted according to the evidence actually obtained.

3.4 Initial Investigation Principle

At this stage, the central distinction is between a statistical trigger and a scientific conclusion. The source-level measurements answer a narrow question:

Which source histories look unusual under the selected statistical representation?

They do not yet answer:

Which sources are astrophysically unusual?

That second question requires evidence from additional views of the observations. The remainder of the methodology is designed around this separation.

4 Evidence-First Investigation Framework

4.1 Scientific Design Principles

The investigation framework was designed around a distinction between statistical unusualness, observational evidence, systematic risk, and astrophysical interpretation. These quantities are related, but they are not interchangeable. A source may be statistically unusual because of an underestimated uncertainty, an instrumental effect, source confusion, or an intrinsically variable astrophysical process. Consequently, a high anomaly score is treated as a reason to investigate rather than as evidence for a new astronomical phenomenon.

The operational discovery chain used in this work is therefore:

UNUSUAL $\rightarrow$ DATA-QUALITY VERIFIED $\rightarrow$ ARTIFACT CHECK
 $\rightarrow$ BLENDING/MOTION CHECK $\rightarrow$ CROSS-SURVEY CONSISTENCY
 $\rightarrow$ KNOWN-CATALOG CHECK $\rightarrow$ ASTROPHYSICALLY PLAUSIBLE
 $\rightarrow$ FOLLOW-UP CANDIDATE.

The final state in this chain is intentionally a candidate state rather than a discovery or confirmation state. Confirmation requires evidence beyond the present exploratory analysis, such as independent observations, additional surveys, or targeted follow-up.

The framework also distinguishes three quantities throughout the investigation:

- **Novelty (N):** how unusual a source appears relative to a defined reference population or model.
- **Evidence quality (E):** how strongly the available observations support the interpretation that the unusual measurement is genuine and source-associated.
- **Systematic risk (R):** how plausible observational, instrumental, processing, or environmental explanations remain.

These quantities are deliberately not collapsed into an arbitrary weighted score. In particular, a source with high statistical novelty but substantial systematic risk should not automatically outrank a source with slightly lower novelty and substantially stronger observational evidence. The resulting prioritization is therefore an investigation decision rather than a claimed probability of astrophysical novelty.

This distinction is important because the same development data are used for exploratory candidate selection, control construction, and diagnostic evaluation. The analysis is consequently presented as a development-field demonstration of an evidence-first workflow rather than as an unbiased estimate of a survey-wide discovery rate.

4.2 Evidence Views and Feature Roles

Each source is represented using multiple evidence views. The views are intended to provide partially complementary descriptions of the source rather than independent measurements.

1. **Temporal view:** repeated measurements, time span, scatter, reduced chi-square, residual structure, influential observations, and leave-one-out stability.
2. **Photometric view:** W1 and W2 behavior, uncertainty-normalized scatter, cross-band relationships, and joint deviations.
3. **Image view:** source position, image-level flux measurements, centroid behavior, morphology, elongation, background properties, and neighboring-source structure.
4. **Observational/systematic view:** quality flags, scan and frame diversity, common-mode behavior, detector or geometric indicators, artifact proximity, and local-control comparisons.
5. **External-survey view:** positional matches, external classifications, variability information, astrometric measurements, and external light curves.
6. **Known-class view:** similarity or discrepancy relative to established variable-source and other known astronomical populations.

The feature representation can be written schematically as

$$\mathbf{x}_s = [\mathbf{x}_{temp}, \mathbf{x}_{phot}, \mathbf{x}_{image}, \mathbf{x}_{sys}, \mathbf{x}_{ext}, \mathbf{x}_{class}],$$

where s denotes a source and each block represents one evidence view.

The views do not all have the same role. Features are assigned one of five roles:

- **DISCOVERY:** useful for identifying unusual sources.
- **EVIDENCE:** useful for determining whether unusual measurements are supported by additional observations.
- **RISK:** useful for identifying possible observational or systematic explanations.
- **CONTEXT:** useful for astrophysical interpretation and comparison with known populations.
- **EVALUATION-ONLY:** useful for auditing the workflow without being allowed to drive candidate selection.

This separation reduces the risk that a variable is used simultaneously as both evidence of an anomaly and evidence that the anomaly detector performed correctly. It does not eliminate dependence between stages, but it makes the provenance of each quantity explicit.

4.3 Reference Populations and Provenance

A central requirement of the framework is that “normal” cannot be defined simply as “not selected by the validator.” A validator-negative source may be genuinely variable but fail the particular operational thresholds, may have insufficient observations, or may differ from the candidate population in ways that make it unsuitable as a normal reference.

The analysis therefore records provenance for reference sources. The principal roles used in the development analysis are:

- **UNVALIDATED_REFERENCE**: sources available for unsupervised modeling but not assigned a variability label by the Phase III validator.
- **VALIDATOR_NEGATIVE_EVALUATION_ONLY**: sources that did not satisfy the validator criteria and are retained for evaluation or diagnostic comparisons, but are not assumed to be normal.
- **VALIDATOR_POSITIVE_EVALUATION_ONLY**: sources that satisfied the operational validator and are used to examine whether later methods recover the selected population.
- **MATCHED_ORDINARY/REFERENCE_CONTROL**: sources selected through local or matched-control procedures for specific comparative tests.

The distinction between these roles is necessary because the available reference populations were constructed internally within the same development field. They are not an independent survey-wide ground truth.

The corrected Phase III audit identified 82 unique NEOWISE groups in the validator-comparison population. Six satisfied the final operational validator criteria. The resulting fraction,

$$\frac{6}{82} = 0.07317,$$

is reported only as an internal comparison-set trigger rate. It is *not* interpreted as a false-positive rate, because the comparison population is not an independently labeled normal population and the validator thresholds were developed within the same analysis.

4.4 Statistical Validator

The first formal reduction from source histories to candidate-like behavior used an uncertainty-aware variability validator. For a source with measurements x_i and corresponding uncertainties σ_i , a weighted mean is used to characterize the source level, while the reduced chi-square provides an operational measure of whether the observed scatter is large relative to the reported uncertainties. The reduced chi-square statistic is

$$\chi^2_\nu = \frac{1}{N-1} \sum_{i=1}^N \frac{(x_i - \mu)^2}{\sigma_i^2},$$

where N is the number of observations and μ is the estimated mean.

The validator used the following operational requirements:

$$N_{\text{obs}} \geq 10,$$

a median W1 signal-to-noise proxy of at least 5, a W1 reduced chi-square of at least 2, and a good-quality fraction of at least 0.90. These thresholds were engineering criteria for candidate generation rather than calibrated statistical significance thresholds.

The threshold audit showed that the number of positive sources was relatively insensitive to the observation-count and signal-to-noise thresholds over the tested alternatives. The dominant reduction occurred when the chi-square requirement was increased: the six-source baseline fell to two sources at $\chi^2_{\nu} \geq 3$ and to zero at $\chi^2_{\nu} \geq 4$. This indicates that the operational candidate population is strongly controlled by the temporal scatter criterion.

The final six validator-positive groups were:

Group	N	Median W1 SNR	W1 χ^2_{ν}	Quality fraction
233	22	35.02	2.118	1.000
234	24	33.41	2.493	1.000
1974	25	22.62	3.026	1.000
1976	25	19.39	2.116	1.000
1978	28	16.45	3.939	1.000
1979	24	14.10	2.279	0.958

The validator should therefore be interpreted as a controlled search-space reduction mechanism. It does not establish that any of these six sources is intrinsically variable.

4.5 Unsupervised Novelty Triage

After construction of the canonical feature matrix, unsupervised methods were introduced to examine whether unusual sources could be identified without relying directly on the validator labels. The first implementation used 15 features available for all 175 source groups. Missing values were median-imputed where necessary, followed by robust scaling. Three complementary unsupervised approaches were examined: Isolation Forest, robust Mahalanobis distance using a minimum covariance determinant estimate, and Local Outlier Factor.

The purpose of this stage was not to construct a classifier or to claim an anomaly probability. Instead, the models were used as exploratory triage mechanisms and as a test of whether the six validator-selected sources occupied unusual regions of feature space for reasons independent of the validator rule.

The three models showed substantial agreement in their overall rankings. The pairwise Spearman rank correlations were 0.8105 for Isolation Forest versus robust Mahalanobis distance, 0.8255 for Isolation Forest versus Local Outlier Factor, and 0.8395 for robust Mahalanobis distance versus Local Outlier Factor. Sixteen of the top twenty sources were shared by all three methods.

The validator-positive sources did not, however, all occupy the highest novelty ranks. Their approximate mean ranks across the three models were:

Group	Isolation Forest	Mahalanobis	LOF	Mean rank
234	8	11	18	12.33
233	17	10	16	14.33
1978	18	19	9	15.33
1974	11	16	20	15.67
1976	25	25	32	27.33
1979	44	45	47	45.33

This result is useful because it demonstrates that the Phase III validator and unsupervised novelty models capture overlapping but non-identical properties. In particular, the system did not simply reproduce the validator labels as an unsupervised ranking.

An important diagnostic emerged from the feature analysis. Brightness and signal-to-noise variables dominated many of the highest novelty ranks. One extreme source, group 93, had a very large W1 reduced chi-square and robust residual, but was not a Phase III survivor. This indicated that raw unsupervised novelty could be strongly influenced by photometric regime rather than by unusual temporal behavior alone.

To investigate this effect, the analysis separated brightness/SNR-related variables from behavior and control variables. Behavioral features were then residualized against the brightness/SNR block before a second Isolation Forest analysis. This reduced the rank correlation between the original and brightness-controlled rankings to 0.6572, demonstrating that brightness was not merely a negligible nuisance variable.

Importantly, groups 1974 and 1978 remained among the stronger sources in the brightness-controlled analysis. Their strongest controlled robust temporal deviations were associated with W1 reduced chi-square rather than with brightness alone. This does not establish astrophysical variability, but it strengthens the interpretation that their unusualness cannot be attributed solely to the simplest brightness/SNR confounding mechanism tested here.

4.6 Feature-Family Diagnostics

The feature space was additionally divided into temporal, sampling, systematic, and photometric families. Each family was independently scaled before unsupervised analysis so that the contribution of one feature family would not automatically dominate because of numerical magnitude or duplication.

The temporal family contained W1 standard deviation, W1 range, W1 reduced chi-square, W1 robust scatter, and W2 standard deviation. The sampling family contained observation count and time span. The systematic family contained positional offsets, quality fraction, and flagged-count measures.

The resulting family-specific ranks showed that groups 1974 and 1978 were consistently strong in the temporal view. Group 1978 ranked 10th in the temporal family, while group 1974 ranked 8th. In contrast, their systematic-family ranks were 58th and 64th, respectively. This difference is informative because it indicates that their overall unusualness was not dominated by the simple systematic proxies available in the canonical source table.

Group 234 provided an important sanity check. It ranked 7th in the temporal family but remained comparatively compatible with matched controls during later evaluation. Thus, a high temporal anomaly measure alone was not treated as sufficient evidence for a robust unresolved candidate.

The feature-family analysis therefore supports a more cautious interpretation: the strongest candidates are sources for which unusual temporal behavior survives several diagnostic reductions, but the feature-family ranks do not provide probabilities of astrophysical origin.

4.7 Robustness and Missingness Handling

Robustness was evaluated at several levels. First, candidate behavior was compared across Isolation Forest, robust Mahalanobis distance, and Local Outlier Factor. Second, Isolation Forest rankings were recomputed over multiple random seeds, population subsamples, and feature subsets. Third, leave-one-out analyses were applied to validator-positive sources with sufficient observations.

The leave-one-out statistic removes one observation at a time and recomputes the reduced chi-square. This provides a direct test of whether an apparent excess is distributed across the source history or is primarily generated by one influential measurement.

For group 1974, all 25 leave-one-out realizations remained above $\chi^2_\nu = 2$, with a minimum of approximately 2.14. Group 1978 showed the strongest robustness, with all 28 leave-one-out realizations remaining above 2 and a minimum of approximately 2.80. Groups 233 and 1976 were more point-sensitive, with minimum leave-one-out values of approximately 1.65 and 1.82, respectively.

The distinction is reflected in the investigation tiers. Groups 1974 and 1978 were treated as higher-priority sources because their temporal excess remained after removal of individual observations. Groups 233 and 1976 were retained as secondary candidates because their validator excess was more sensitive to individual measurements.

Missing evidence was handled separately from negative evidence. If image measurements, external catalogue matches, or known-class information were unavailable, the corresponding state was recorded as UNKNOWN rather than NORMAL or NEGATIVE. This distinction is particularly important for external surveys: a failed Gaia query cannot be interpreted as evidence that no Gaia counterpart exists, and the absence of a valid SIMBAD return cannot be interpreted as proof that the source has no astronomical identity.

The same principle applies to image and artifact data. Failure to retrieve a mask or metadata table is not evidence that the corresponding artifact is absent. Instead, the evidence state remains incomplete.

This approach makes the resulting candidate dossiers deliberately asymmetric. A source can have strong temporal evidence while simultaneously having incomplete external evidence and unresolved

systematic risk. Such a source remains a candidate for further investigation rather than being forced into a binary “real” or “artifact” classification.

5 Investigation Results

5.1 Search-Space Reduction

The investigation began with 8,652 retrieved NEOWISE single-exposure detections from the development field. After requiring usable W1 measurements, 7,308 detections remained. Quality information was then used to characterize the observations rather than simply discarding all flagged measurements: 2,498 detections satisfied the selected quality condition, while 4,810 were flagged. The initially selected development candidate was removed from the population before constructing the final source histories, leaving 7,275 working detections.

Grouping detections into source-like histories produced 3,371 source groups. Of these, 175 contained at least ten observations and therefore satisfied the minimum temporal sampling requirement used by the operational validator.

Stage	Number
Retrieved detections	8,652
Usable W1 detections	7,308
Working detections after candidate removal	7,275
Source-like groups	3,371
Groups with at least 10 observations	175
Corrected validator-comparison groups	82
Validator-positive groups	6

This reduction illustrates the intended role of the pipeline. The majority of detections are not individually treated as anomaly candidates. Instead, repeated detections are consolidated into source histories, and only sources with sufficient temporal information are passed to the variability-oriented screening stage.

The reduction should not be interpreted as an estimate of survey completeness. Several stages intentionally remove observations or restrict the analysis to a small development field, and the source grouping itself can depend on positional association. The resulting population is therefore an analysis population rather than a representative sample of all NEOWISE sources.

5.2 Validator Results and Threshold Sensitivity

The corrected validator identified six positive source groups among 82 unique comparison groups. The six sources were groups 233, 234, 1974, 1976, 1978, and 1979. Every source satisfied the baseline requirements of at least ten observations, median W1 signal-to-noise proxy of at least five, W1 reduced chi-square of at least two, and good-quality fraction of at least 0.90.

The validator-positive fraction in this comparison set was

$$\frac{6}{82} = 7.32\%.$$

This value is deliberately not described as a false-positive rate. The 82 sources were selected from the same development analysis and were not independently labeled as non-variable sources. Some may contain genuine variability, and some may be affected by effects that are not represented by the validator features. The quantity is therefore an internal trigger fraction only.

Threshold sensitivity was examined to determine which operational criteria controlled the resulting candidate set.

Threshold modification	Positive groups	Change from baseline
Baseline	6	–
$N_{\text{obs}} \geq 15$	6	0
$N_{\text{obs}} \geq 20$	6	0
Median W1 SNR ≥ 7	6	0
Median W1 SNR ≥ 10	6	0
$\chi^2_{\nu} \geq 3$	2	-4
$\chi^2_{\nu} \geq 4$	0	-6
Good-quality fraction ≥ 0.95	6	0
Good-quality fraction = 1	5	-1

The dominant sensitivity is therefore associated with the W1 reduced chi-square threshold. This is expected because the validator was designed primarily to detect excess temporal scatter relative to the reported photometric uncertainties. It also demonstrates why the validator cannot be treated as an independent confirmation step: the final candidate population is strongly determined by an explicitly chosen operational statistic.

The six sources have W1 reduced chi-square values ranging from approximately 2.12 to 3.94. Group 1978 is the strongest of the six by this metric, while groups 233 and 1976 are only slightly above the baseline threshold.

5.3 Candidate Robustness

A central question was whether the observed temporal excess was distributed across the source history or generated primarily by one influential measurement. Leave-one-out analysis was therefore applied to the four candidates with complete saved leave-one-out provenance: groups 233, 1974, 1976, and 1978.

For each source, one observation was removed and the reduced chi-square was recomputed. The minimum and maximum values provide a compact description of the sensitivity of the result to individual observations.

Group	N	Full χ^2_ν	Minimum LOO	Maximum LOO
233	22	2.118	1.653	2.224
1974	25	3.026	2.142	3.157
1976	25	2.116	1.824	2.207
1978	28	3.939	2.798	4.091

Groups 1974 and 1978 show the strongest leave-one-out robustness. Every individual removal leaves the reduced chi-square above two. This suggests that their excess scatter is distributed across multiple observations rather than being completely explained by a single point.

Groups 233 and 1976 are less robust under the same test. Approximately 90.9 percent of the leave-one-out realizations for group 233 and 88 percent for group 1976 remain at or above the baseline threshold. These sources therefore remain useful secondary candidates but do not receive the same priority as groups 1974 and 1978.

The influential-observation analysis further supports this distinction. The strongest full-history standardized residuals were approximately 3.30 for group 233, 4.77 for group 1974, 2.90 for group 1976, and 5.69 for group 1978. The corresponding leave-one-out reductions were sufficient to move groups 233 and 1976 below the baseline threshold in at least some realizations, whereas groups 1974 and 1978 remained above it.

This result is not evidence that the measurements are astrophysical. A systematic effect can also affect multiple observations. Leave-one-out stability therefore establishes persistence of the measured statistical property, not persistence of an astrophysical phenomenon.

5.4 Unsupervised Model Agreement and Stability

The unsupervised analysis was designed to test whether the candidate set remained unusual under alternative descriptions of the source population. Three model families were compared: Isolation Forest, robust Mahalanobis distance, and Local Outlier Factor.

The pairwise rank correlations were:

Model pair	Spearman ρ
Isolation Forest–Mahalanobis	0.8105
Isolation Forest–LOF	0.8255
Mahalanobis–LOF	0.8395

Sixteen of the top twenty sources were shared by all three models. This level of agreement indicates that a substantial component of the detected structure is not specific to one anomaly-detection algorithm.

However, model agreement does not constitute validation against a known truth set. In addition, exact candidate ranks varied under changes to the random seed, population subsampling, and feature subsets. The median seed stability of the Isolation Forest was high at the population

level, with a median rank correlation of approximately 0.99 across tested seeds, but candidate-level top-ten membership was substantially less stable.

For example, group 1978 repeatedly occupied an unusual region of feature space but did not consistently retain the same exact rank. Group 1974 was somewhat more rank-stable across the tested perturbations. These results motivate the use of broad investigation tiers rather than treating a particular machine-learning rank as a precise measurement.

No accuracy, precision, recall, F1 score, area under the ROC curve, or false-positive rate is reported for these models because no independent ground-truth anomaly labels were available.

5.5 Brightness and Signal-to-Noise Diagnostics

Anomaly detection in astronomical photometry can be strongly influenced by observational regime. Bright sources may have small formal uncertainties, while faint sources can display larger normalized or absolute scatter. If brightness and signal-to-noise variables dominate an unsupervised model, a high anomaly score may describe a photometric regime rather than unusual source behavior.

This effect was directly tested. Four variables associated primarily with brightness and signal-to-noise were separated from a behavioral and control feature block. The behavioral block included W1 scatter, W1 range, W1 reduced chi-square, W1 robust scatter, W2 scatter, temporal coverage, observation count, positional offsets, quality fraction, and flagged-count measures.

After robust residualization of the behavioral variables against the brightness/SNR block, the rank correlation between the original and controlled Isolation Forest rankings was 0.6572. The reduction indicates that brightness-related information was a substantial component of the original novelty structure.

The test nevertheless retained groups 1974 and 1978 among the more unusual sources. Their strongest controlled deviations remained associated with temporal behavior, particularly W1 reduced chi-square. This provides evidence against the narrow interpretation that their ranking was caused solely by brightness or signal-to-noise.

The analysis also showed that candidate behavior was heterogeneous. Group 1978 retained a strong controlled temporal signal, whereas groups 233 and 1976 became much less prominent after the brightness/SNR correction. Group 234 remained an important sanity-check source because it could be highly ranked in temporal feature space while remaining comparatively compatible with matched controls.

These results support the use of brightness/SNR-controlled diagnostics as an audit of the anomaly model rather than as a new candidate score.

5.6 Feature-Family Comparison

To identify which types of measurements were driving the unsupervised results, the canonical feature matrix was divided into temporal, sampling, systematic, and photometric families.

The temporal family included W1 standard deviation, W1 range, W1 reduced chi-square, W1 robust scatter, and W2 standard deviation. The sampling family contained observation count and

time span. The systematic family contained positional offsets, quality fraction, and flagged-count measures.

The candidate ranks obtained from these independent families were:

Group	Temporal	Sampling	Systematic	Photometric
233	14	166	39	19
234	7	113	49	11
1974	8	101	64	15
1976	31	101	52	10
1978	10	52	58	22
1979	43	87	21	24

The temporal family produced the strongest candidate separation for groups 1974 and 1978. Their systematic-family ranks were substantially less unusual. This pattern is consistent with the interpretation that their initial selection was driven more by temporal photometric behavior than by the simple positional and quality proxies included in the canonical source table.

The family rankings also demonstrate why the final investigation cannot be reduced to a single anomaly score. Different feature families answer different questions. Temporal features ask whether the source history is unusual; systematic features ask whether observational structure accompanies the unusualness; photometric features examine relationships between bands; and sampling features describe how the source was observed.

The Spearman correlations between the temporal, sampling, systematic, and photometric family rankings were also heterogeneous. Temporal and sampling ranks were nearly uncorrelated ($\rho = 0.013$), while systematic and photometric ranks showed a stronger association ($\rho = 0.501$). This heterogeneity supports retaining multiple evidence views rather than assuming that one scalar ranking captures the complete observational state of a source.

5.7 Multi-View Evidence Profiles

The six validator-positive sources were evaluated using four currently populated views: temporal, sampling, systematic, and photometric. External-catalogue and known-class views remained unavailable for the final candidate set because the relevant external information could not be established consistently.

Group	Temporal	Sampling	Systematic	Photometric	Evidence state
233	14	166	39	19	LIMITED
234	7	113	49	11	LIMITED
1974	8	101	64	15	HIGHER PRIORITY
1976	31	101	52	10	LIMITED
1978	10	52	58	22	HIGHER PRIORITY
1979	43	87	21	24	LIMITED

Groups 1974 and 1978 combine strong temporal unusualness with relatively unexceptional systematic-family ranks. Both also survive the leave-one-out analysis. This combination motivated their promotion to the highest investigation tier.

The remaining four sources were retained for different reasons. Groups 233 and 1976 showed validator-level temporal excess but weaker leave-one-out robustness. Groups 234 and 1979 were retained as sanity controls because their statistical behavior could resemble candidate behavior while later matched-control comparisons did not provide the same level of support.

The multi-view analysis does not assign numerical probabilities to these states. In particular, the relative ranks in the table are descriptive outputs from separate analyses and are not combined through an arbitrary weighted sum. Candidate priority is instead determined from the accumulated evidence and unresolved risks documented in the subsequent observational and astrophysical vetting.

6 Observational and Astrophysical Vetting

6.1 Image-Level Evidence

Statistical analysis alone cannot determine whether an unusual photometric measurement originates from the intended astronomical source. Image-level inspection was therefore introduced as an independent diagnostic stage for the highest-priority candidates.

The image investigation used official NEOWISE image products and WCS information to identify the target position in individual W1 and W2 frames. For the two highest-priority candidates, groups 1974 and 1978, influential observations identified by the leave-one-out analysis were compared with observations in which the source showed more typical photometric behavior.

The image photometry used a WCS-anchored aperture centered on the target position. A three-pixel aperture was combined with a background annulus extending from six to twelve pixels. Robust background estimates were obtained using the median and median absolute deviation (MAD). This procedure was used as an independent diagnostic rather than as a replacement for the calibrated NEOWISE catalogue photometry.

For group 1974, the median aperture flux of the four influential observations was approximately 597.1 in the image-flux units used by the extraction, compared with approximately 643.1 for the three selected comparison observations. The corresponding median signal-to-noise values were approximately 24.4 and 25.2. For group 1978, the influential observations had a median flux of approximately 453.7 compared with approximately 414.2 for the comparison observations, with median signal-to-noise values of approximately 21.2 and 19.9.

The differences therefore do not have a single common direction across the two candidates. This is important because image-level flux differences can result from extraction, background, point-spread-function, pixel-sampling, or other observational effects. The image measurements establish that source-like signal is spatially present at the expected position, but they do not establish that the source itself underwent intrinsic astrophysical variability.

A further bookkeeping discrepancy was identified during the image analysis: finite-pixel handling could alter extracted fluxes by integer multiples of the adopted background contribution. The discrepancy was documented rather than silently corrected. Consequently, the image measurements are interpreted primarily through source presence, relative morphology, and consistency of the observational configuration rather than as a second independently calibrated light curve.

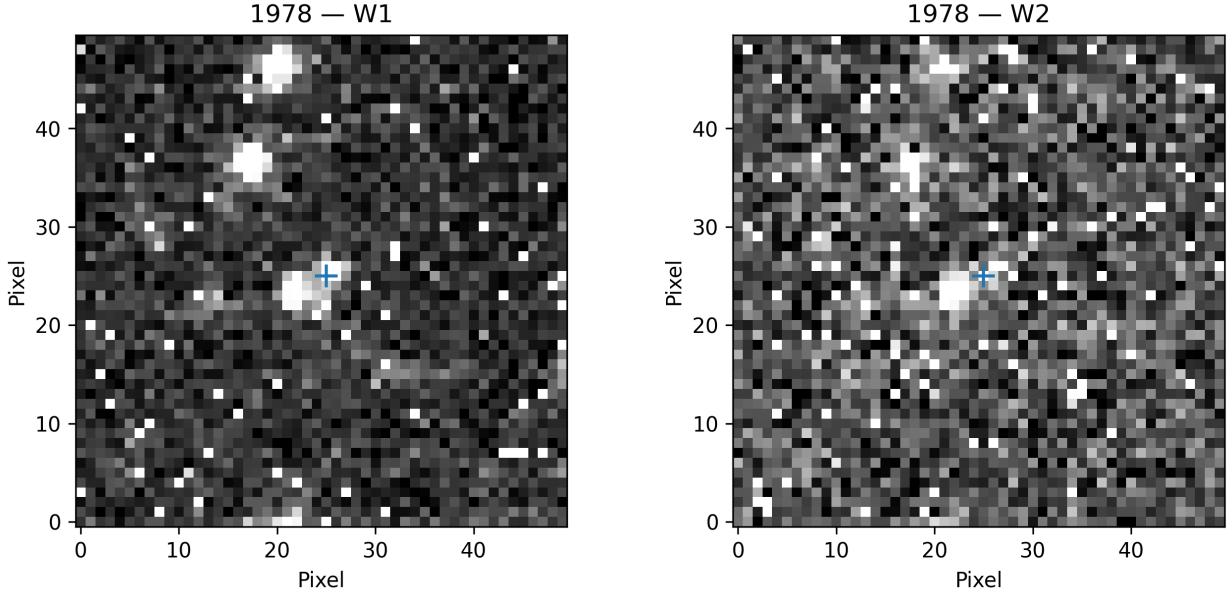


Figure 1: Image-level inspection of candidate source 1978 in an influential observation used in the robustness analysis. The W1 and W2 panels are centered on the WCS-derived source position. The source is spatially present in the W1 image, while the corresponding W2 signal is weaker relative to the local background. These images provide image-level evidence that the measured source is present in the observation, but they do not by themselves establish intrinsic astrophysical variability.

The direct image comparison also identified a notable W1 elongation for group 1978 in one influential observation. This morphology is potentially relevant to source confusion or extraction effects, but morphology alone is not sufficient to assign a cause. It is therefore retained as evidence requiring interpretation rather than classified as either an artifact or an astrophysical signal.

6.2 Artifact and Calibration Checks

NEOWISE single-exposure images can contain several classes of observational structure, including diffraction spikes, scattered light, latent images, optical ghosts, charged-particle effects, electronic noise, and other image-level artifacts. The released source products also contain detections that may be spurious or affected by artifacts. Consequently, artifact inspection was treated as a required stage before assigning high priority to a candidate.

For the 16 inspected W1/W2 image cutouts associated with the most influential observations of groups 1974 and 1978, artifact-table metadata were examined for diffraction spikes, halos, latents,

and optical ghosts. Diffraction-spike entries were present for 13 of the 16 inspected images and absent from three. Halo entries were present in all 16. Latent entries were present in 12 and absent in four. Optical-ghost entries were present in three images and absent in 13.

The nearest listed artifact locations were generally hundreds of arcseconds from the target positions, approximately 383–400 arcsec in the inspected records. No nearby listed artifact was identified within the local search radii examined in this audit.

This result is useful but must be stated narrowly. A missing artifact-table entry does not prove that an image is artifact-free, and the mask-download portion of the audit was incomplete. The appropriate conclusion is therefore that no nearby listed artifact was identified in the inspected metadata, rather than that all image artifacts were excluded.

Photometric calibration was considered separately from image artifacts. The global W1 calibration of NEOWISE is substantially more stable than the roughly tenth-of-a-magnitude-scale scatter seen in the strongest candidate histories, making a simple global W1 zero-point drift an insufficient explanation by itself. However, global calibration stability does not exclude local detector, extraction, background, or scan-dependent effects.

W2 was treated conservatively because its observational behavior includes stronger seasonal structure than W1. In the present candidates, however, W2 did not reproduce the same level of excess scatter as W1. This asymmetry is informative but inconclusive: it does not prove that W1 is astrophysical, nor does it require W2 to display identical behavior.

The distinction between aperture photometry and profile-fit photometry was also retained. Differences between these measurement approaches can vary with observing conditions and source environment. Therefore, agreement or disagreement between catalogue and image-level measurements is interpreted as a diagnostic of the measurement process rather than as a direct test of physical variability.

6.3 Blending and Local-Neighbor Checks

Nearby-source contamination is one of the principal alternative explanations for unusual aperture photometry. A source that appears isolated in a catalogue may nevertheless receive flux from a nearby object, particularly when the point-spread function or extraction aperture changes relative to the source geometry.

Local NEOWISE source histories were therefore examined within a 30-arcsec neighborhood around the highest-priority candidates. Group 1978 had two nearby source groups: group 1977 at approximately 11.29 arcsec and group 2054 at approximately 18.69 arcsec. Group 1974 had group 1975 at approximately 28.04 arcsec.

The nearest-neighbor histories were compared with the candidate histories. For group 1978, the nearest neighbor, group 1977, showed a correlation of approximately -0.082 with the target W1 measurements. For group 1974, group 1975 showed a correlation of approximately 0.148.

These weak correlations do not provide strong evidence that the candidate light curves are simply reproducing the temporal behavior of the nearest neighbor. The result is therefore classified as

weak or inconclusive evidence against a simple neighbor-variability explanation.

However, this test does not eliminate blending. Contamination can occur without the neighboring source displaying measurable variability, and the relevant point-spread-function footprint need not map directly onto the nearest catalogue source. The existence of nearby sources therefore remains an unresolved systematic risk, especially for group 1978.

The analysis deliberately avoids interpreting the neighbor test as a binary blend/no-blend classifier. Instead, it establishes that the simplest form of correlated neighbor variability was not strongly observed in the tested histories while retaining source confusion as a plausible alternative explanation.

6.4 Same-Scan, Frame, and Local Systematic Checks

Observations obtained during the same scan or frame can share observational conditions. If several candidates exhibit unusual measurements under the same configuration, a common systematic process becomes a plausible explanation.

Earlier control analyses examined shared scan identifiers, frame identifiers, and matched local observations. No dominant common mode was identified among the strongest candidates. In the matched same-scan/frame local-control analysis, the candidate-control correlations were modest, including approximately 0.128 for group 1974, 0.072 for group 1976, 0.125 for group 1978, and 0.288 for group 1979.

The scan-level analysis was limited by the available comparison observations, while the frame-level analysis provided only a small number of usable correlations. Consequently, the absence of a dominant common pattern should not be interpreted as elimination of scan- or detector-dependent effects.

A positional-proxy analysis likewise did not identify a dominant geometry signature shared by the strongest candidates. Detector-level pixel phase or physical detector coordinates were not directly measured in this analysis, so these mechanisms remain unresolved.

The observational evidence therefore converges on a cautious position: several simple common-mode explanations were not strongly supported by the available controls, but the analysis does not establish that all instrumental or extraction-related explanations have been excluded.

6.5 Cross-Band Photometric Behavior

The relationship between W1 and W2 provides an additional diagnostic because a genuinely astrophysical process may produce correlated behavior across infrared bands, whereas some measurement effects can affect the bands differently.

For group 1978, the W1 reduced chi-square from the principal candidate analysis was approximately 3.94, while the W2 reduced chi-square was approximately 1.29. The W1 standard deviation was approximately 0.137 mag and the W2 standard deviation approximately 0.250 mag. The W1–W2 correlation was approximately 0.303.

For group 1974, the corresponding W1 reduced chi-square was approximately 3.03 and W2 approximately 1.17. The W1 standard deviation was approximately 0.099 mag and the W2 standard deviation approximately 0.129 mag, with a W1–W2 correlation of approximately 0.041.

These measurements show that neither candidate has a simple, strong, broadband-correlated variability signature in the available data. In particular, W2 does not reproduce the W1 excess in reduced chi-square.

The result is therefore treated as asymmetric evidence. It prevents the W1 behavior from being described as independently confirmed across both NEOWISE bands. At the same time, the weaker W2 behavior does not by itself invalidate the W1 measurements, because band-dependent noise, seasonal effects, sensitivity, and source spectral behavior can differ.

The cross-band analysis consequently supports an “inconclusive” rather than a positive or negative astrophysical interpretation.

6.6 External-Catalogue Checks

External catalogues were queried to determine whether the highest-priority candidates already possessed known astronomical identities or classifications. SIMBAD and Gaia were treated as complementary resources, with query failures explicitly distinguished from non-matches.

For groups 1974 and 1978, the SIMBAD queries returned no valid astronomical object for the tested coordinates. This result means only that no valid SIMBAD object was returned under the query conditions. It does not establish that the sources lack an astronomical identity.

The Gaia queries encountered retrieval failures in the analysis environment. These failures were therefore recorded as UNKNOWN rather than interpreted as evidence that no Gaia counterpart exists.

Earlier experiments using external catalogues demonstrated the same general limitation. A small GCVS comparison identified one known RR Lyrae-type object in the development field, while a Gaia variable-class comparison returned only a small number of classified sources. These results were useful as feasibility checks but were far too small to constitute an independent benchmark.

Consequently, the external-catalogue stage did not establish a known class for the Tier 1 candidates. The correct scientific state is therefore “external identity unresolved”, not “new object” or “unknown astronomical class”.

7 Final Candidate Dossiers

7.1 Candidate 1978

Group 1978 is the highest-priority source produced by the combined investigation. It contains 28 observations and has a baseline W1 reduced chi-square of approximately 3.94, a median W1 signal-to-noise proxy of approximately 16.45, and a good-quality fraction of 1.00. Its leave-one-out reduced chi-square remains above two for every tested removal, with a minimum of approximately 2.80 and a maximum of approximately 4.09.

The robustness result is particularly important because the candidate is not selected solely because of one extreme measurement. The temporal excess remains present when individual observations are removed. This makes the source a stronger investigation target than candidates whose validator status depends heavily on one observation.

The W2 measurements do not reproduce the same strength of excess scatter. The principal analysis gives a W2 reduced chi-square of approximately 1.29 and a W1–W2 correlation of approximately 0.30. Thus, the available broadband evidence is inconclusive rather than independently confirmatory.

Image-level inspection establishes source-like signal at the expected WCS position. The influential observations also show measurable differences in W1 image properties, including a notable elongation in one observation. However, these measurements cannot distinguish intrinsic variability from changes in point-spread-function sampling, extraction, background, or source confusion.

Two nearby NEOWISE source groups were identified within 30 arcsec, including group 1977 at approximately 11.29 arcsec and group 2054 at approximately 18.69 arcsec. The nearest-neighbor tracking test did not reveal strong correlated variability with group 1977. The measured correlation was approximately -0.082 . This weakens a simple explanation in which the target merely follows the measured variability of its nearest neighbor, but it does not eliminate aperture contamination or unresolved blending.

No valid SIMBAD object was returned under the tested query conditions, while the Gaia query remained unresolved because of a retrieval failure. The external identity of the source must therefore be classified as UNKNOWN.

Taken together, the evidence supports the following conservative statement: group 1978 is a high-priority candidate whose W1 temporal behavior is unusually strong within the development analysis and remains robust under individual-observation removal, while several simple systematic explanations were not strongly supported by the available tests. Nevertheless, blending, extraction effects, detector-level systematics, incomplete artifact-mask information, and the lack of independent external confirmation prevent the source from being classified as a confirmed astrophysical variable or a new astronomical phenomenon.

7.2 Candidate 1974

Group 1974 is the second highest-priority candidate. It contains 25 observations, has a baseline W1 reduced chi-square of approximately 3.03, a median W1 signal-to-noise proxy of approximately 22.62, and a good-quality fraction of 1.00.

The leave-one-out analysis provides strong robustness. All individual observation removals retain a W1 reduced chi-square above two, with a minimum of approximately 2.14 and a maximum of approximately 3.16. The observed temporal excess is therefore distributed across the source history rather than being entirely attributable to one influential measurement.

The W2 reduced chi-square is approximately 1.17, while the W1–W2 correlation is only approximately 0.041. As with group 1978, the cross-band evidence does not independently reproduce the

W1 behavior.

Image-level photometry gives a median flux of approximately 597.1 for the selected influential observations and approximately 643.1 for the selected comparison observations. The corresponding median signal-to-noise values are approximately 24.4 and 25.2. These differences demonstrate that the source and local measurement environment differ between observations, but they do not establish the physical origin of the difference.

One nearby NEOWISE source, group 1975, was identified at approximately 28.04 arcsec. Its measured W1 behavior showed only weak correlation with the target, approximately 0.148. As for group 1978, this result does not eliminate blending because a contaminating source need not itself be variable.

The external identity of group 1974 also remains unresolved. No valid SIMBAD object was returned in the tested query and the Gaia query failed at the retrieval stage.

Group 1974 is consequently classified as a high-priority unresolved candidate. Its temporal behavior is robust within the tested analysis and is unusual relative to the matched controls, but the available observations do not justify a claim of intrinsic astrophysical variability.

7.3 Secondary Candidates 233 and 1976

Groups 233 and 1976 satisfy the same baseline validator requirements but receive lower investigation priority because their temporal evidence is less robust to individual observations.

Group 233 has 22 observations and a W1 reduced chi-square of approximately 2.12. Its minimum leave-one-out value is approximately 1.65, meaning that removal of some individual observations reduces the statistic below the baseline validator threshold. The strongest full-history standardized residual is approximately 3.30. This source therefore remains interesting, but its candidate status is more sensitive to individual measurements.

Group 1976 has 25 observations and a W1 reduced chi-square of approximately 2.12. Its minimum leave-one-out value is approximately 1.82, and its strongest full-history standardized residual is approximately 2.90. Like group 233, it retains evidence of excess scatter but does not demonstrate the same distributed robustness as groups 1974 and 1978.

Both sources are therefore classified as Tier 2 candidates. Their status is best described as requiring additional controls and independent observations rather than as evidence of an established astrophysical phenomenon.

7.4 Sanity Controls 234 and 1979

Groups 234 and 1979 were retained as important sanity checks rather than promoted to the highest candidate tier.

Group 234 has a W1 reduced chi-square of approximately 2.49 and a median W1 signal-to-noise proxy of approximately 33.41. It ranks highly in several unsupervised analyses, including the temporal feature family. Nevertheless, matched-control comparisons did not provide the same level of support as for groups 1974 and 1978.

Group 1979 has a W1 reduced chi-square of approximately 2.28 and a median W1 signal-to-noise proxy of approximately 14.10. Its good-quality fraction is approximately 0.958. Its systematic-family rank is comparatively unusual, while its temporal-family rank is weaker than those of the principal Tier 1 candidates.

These sources demonstrate why anomaly score, validator status, and candidate priority must remain separate. A source can satisfy an operational statistical criterion and still fail to accumulate enough independent evidence for high-priority follow-up.

The final investigation tiers are therefore:

Group	Tier	Interpretation
1978	TIER 1 – HIGHEST	High-priority unresolved candidate
1974	TIER 1 – HIGH	High-priority unresolved candidate
233	TIER 2	Secondary candidate
1976	TIER 2	Secondary candidate
234	CONTROL/SANITY	Control-compatible sanity source
1979	CONTROL/SANITY	Control-compatible sanity source

No source in the final candidate set is classified as confirmed.

8 Discussion

8.1 What the Analysis Establishes

The principal result of this work is methodological rather than the discovery of a new astronomical phenomenon. In a small NEOWISE development field, an evidence-first workflow was able to reduce a large detection population to a small set of sources requiring progressively more detailed investigation. No formal multiple-testing-adjusted discovery significance is claimed. The candidate tiers are investigative priorities within the development analysis, not survey-wide statistical discoveries.

The workflow demonstrated that statistical unusualness can be separated from subsequent observational vetting. Six sources passed the operational variability validator, but only two, groups 1978 and 1974, accumulated sufficient robustness and multi-view evidence to become Tier 1 investigation targets.

The leave-one-out analysis was particularly informative. It separated candidates with distributed temporal excess from candidates whose validator status was more dependent on individual observations. The unsupervised analysis provided a second perspective and showed that the validator-selected sources were not simply reproduced by a single machine-learning rule.

The feature-family analysis further showed that temporal behavior was a stronger contributor to the unusualness of groups 1974 and 1978 than the simple systematic proxies available in the canonical source matrix. Brightness and signal-to-noise effects were also demonstrated to influence

unsupervised rankings, providing a concrete example of why astronomical anomaly detection should not interpret an unsupervised score without examining its feature drivers.

The observational investigation produced a similarly mixed result. No dominant shared scan or frame pattern was identified, no nearby listed artifact was identified in the inspected artifact meta-data, and simple nearest-neighbor variability did not strongly reproduce the candidate behavior. At the same time, nearby sources remain present, image-level extraction effects remain possible, detector-level effects were not directly measured, and the artifact-mask audit was incomplete.

The final result is therefore not “two new variables.” It is more accurately stated as two high-priority unresolved candidates whose measured W1 behavior survived several layers of internal robustness and systematic testing but remains unconfirmed.

8.2 The Importance of the W1/W2 Asymmetry

The different behavior of W1 and W2 is one of the most important cautionary observations in the analysis.

For group 1978, the W1 reduced chi-square is approximately 3.94 while the W2 value is approximately 1.29. For group 1974, the corresponding values are approximately 3.03 and 1.17. The candidates therefore do not exhibit a simple broadband pattern in which both bands independently reproduce the same statistical excess.

This asymmetry has several possible interpretations. The W1 measurements could contain an astrophysical signal that is weaker or absent in W2, the two bands could have different measurement uncertainties and systematic behavior, or an observational process could affect the bands differently. The present analysis does not contain enough independent information to distinguish these possibilities.

Consequently, W2 disagreement is retained as a limitation rather than selectively ignored. An evidence-first analysis should allow evidence to weaken a candidate interpretation even when other measurements appear favorable.

8.3 Blending as an Unresolved Alternative

Source confusion remains one of the most important unresolved alternatives for the Tier 1 candidates.

The presence of nearby sources means that catalogue-level photometry cannot automatically be interpreted as the isolated flux of a single astronomical object. The local-neighbor analysis did not find strong correlated variability in the nearest tested sources, but this is not equivalent to proving that no contaminating flux enters the aperture.

A contaminating source could be approximately constant while changes in the target’s position, point-spread function, background, or extraction geometry alter the fraction of contaminating light measured by the aperture. Conversely, a nearby source could be outside the simple neighbor-tracking test while still contributing to the effective photometric footprint.

For this reason, the appropriate conclusion is “blending weak or inconclusive” rather than “blending rejected.”

8.4 Relationship to Known Variable-Source Catalogues

The external catalogue experiments provide a useful but limited context for the candidates.

A small GCVS test identified a known RR Lyrae-type source in the development field. Gaia-based tests likewise demonstrated that some known variable classes could be recovered under the available cross-match procedure. However, these samples were too small to establish classification performance or completeness.

The Gaia variable-class experiment recognized one of three tested example classes in the small available benchmark. This result should not be interpreted as a classifier accuracy measurement. The external classes were used as recognition and sanity checks, not as a comprehensive ground-truth dataset.

For the Tier 1 candidates, the absence of a valid SIMBAD return and the failure of the Gaia query leave the external identity unresolved. This prevents a strong claim that either source is absent from known catalogues.

The correct scientific position is therefore that the current analysis has not established a known classification for the Tier 1 candidates, rather than that it has established a previously unknown class.

8.5 Interpretation of the Machine-Learning Stage

The machine-learning stage should be understood as exploratory triage.

Isolation Forest, robust Mahalanobis distance, and Local Outlier Factor showed substantial rank agreement, but their output was affected by brightness and signal-to-noise features. The brightness-controlled analysis reduced this effect and retained groups 1974 and 1978 among the stronger temporal outliers.

The stability experiments also showed an important distinction between population-level model stability and candidate-level rank stability. A model can be globally stable while individual sources move substantially in rank under population or feature perturbations.

For this reason, the analysis does not assign a probability to any candidate based on its anomaly score. Nor does it report classification metrics that require an independent ground-truth label set. The machine-learning contribution is instead the demonstration that multiple unsupervised descriptions can help identify unusual regions of feature space while exposing confounding structure that must subsequently be investigated.

8.6 Candidate Priority Is Not a Probability

The final candidate tiers should not be interpreted as a calibrated probability that a source is astrophysically unusual.

The Tier 1 designation reflects an accumulation of evidence: strong temporal behavior, leave-one-out robustness, matched-control unusualness, relatively weak simple systematic signatures, and the absence of a satisfactory explanation among the tested alternatives. It also reflects unresolved risks, including nearby sources and incomplete external information.

This is fundamentally different from assigning a numerical probability such as “80 percent likely to be astrophysical.” Such a probability would require a calibrated generative or classification framework and a representative validation population, neither of which is available here.

The tier system therefore serves an operational purpose: it tells a researcher where additional effort is most justified.

8.7 No Claim of New Physics or a New Astronomical Class

Nothing in the present analysis establishes a new physical mechanism, new astronomical class, or anomalous object population.

The word “unexplained” is used only in the restricted sense that the source behavior remained unresolved after the available tests. It does not mean that all possible conventional explanations have been eliminated.

In particular, the Tier 1 candidates remain compatible with several unresolved possibilities, including measurement-systematic effects, source confusion, extraction effects, or ordinary astrophysical variability that has not yet been identified through the available external information.

A scientifically useful anomaly investigation therefore ends the current analysis with a follow-up recommendation rather than a discovery declaration.

9 Limitations

Several limitations materially constrain the interpretation of the results.

9.1 Development-Field Selection

The analysis was performed in a small development field centered near $RA \approx 180^\circ$ and $Dec \approx 0^\circ$. The field was selected for methodological development rather than as a representative random sample of the NEOWISE sky.

Consequently, the observed candidate fractions, feature distributions, and systematic behavior cannot be generalized to the full survey. The six validator-positive sources among the 82 comparison groups are an internal development-set result, not a survey-wide incidence estimate.

9.2 Non-Independent Candidate Selection and Evaluation

Candidate selection, control construction, and several diagnostic evaluations were performed using related data from the same development field. This creates opportunities for researcher degrees of freedom and selection effects.

The analysis attempts to mitigate these effects through explicit provenance, threshold sensitivity, multiple diagnostic methods, retained controls, rejected candidates, and a documented analysis freeze. These measures improve transparency but do not transform the development analysis into an independent statistical validation.

A future study should evaluate the frozen pipeline on an independently selected field or, preferably, on a predefined survey-wide test population that was not used during method development.

9.3 Operational Thresholds

The validator thresholds were chosen as engineering criteria for search-space reduction. They were not derived from a formal false-discovery-rate calibration or a predefined probabilistic model.

The sensitivity analysis showed that the W1 reduced chi-square threshold dominates the resulting candidate population. This dependence means that small changes to the operational threshold can substantially change the number of selected candidates.

The thresholds should therefore be regarded as reproducible settings of the present development pipeline rather than universal definitions of astronomical variability.

9.4 Multiple Testing

The investigation examined many sources and many features. Formal hypothesis-testing corrections for the complete exploratory search were not performed, and no survey-wide discovery significance or p-value is claimed.

This is intentional. The present work treats statistical unusualness as a triage signal and uses observational vetting to determine whether a source deserves additional attention. A future confirmatory analysis would require a predefined search space, frozen selection criteria, explicit multiplicity control, and an independent validation sample.

9.5 Reference-Population Limitations

The available control populations are internally constructed and comparatively small. Some controls may themselves contain genuine variability or unrecognized systematic effects.

The matched-control experiments are therefore interpreted as comparative diagnostics rather than ground truth. A source exceeding all ten matched controls, for example, is described as unusual relative to those ten controls rather than statistically significant at a particular confidence level.

The same limitation applies to the 82-source validator comparison set. It cannot provide an unbiased estimate of the validator’s false-positive rate.

9.6 External-Data Incompleteness

External catalogue information is incomplete for the final candidates. SIMBAD returned no valid object under the tested conditions, while Gaia retrieval failed in the analysis environment.

The correct representation of these states is UNKNOWN. Treating failed queries or missing classifications as negative evidence would artificially strengthen the candidate interpretation.

Future work should repeat the external cross-matching using a robust, independently validated data-access environment and include additional catalogues and survey-specific light curves where available.

9.7 Image and Instrumental Limitations

Image-level inspection was performed for selected influential observations rather than every detection. The image analysis therefore provides targeted diagnostic evidence rather than a complete image-level audit of each source history.

The artifact metadata audit identified no nearby listed artifact in the inspected region, but the mask-download portion remained incomplete. Detector pixel-phase effects and detailed detector-coordinate systematics were not directly measured.

The image-level aperture photometry was also an independent diagnostic rather than a replacement for the calibrated NEOWISE catalogue measurements. Differences between the two should not be interpreted as a direct measurement of astrophysical variability.

9.8 Temporal Sampling

Although the source histories contain many individual detections, the observations are not uniformly distributed in time. Several diagnostic analyses necessarily relied on a small number of broad temporal epochs.

Accordingly, statements about temporal persistence refer primarily to robustness of the measured scatter under observation removal and to repeated observations, rather than to a detailed characterization of a periodic or long-term astrophysical light curve.

A genuine characterization of periodicity, transient behavior, or long-term evolution would require more appropriate time-series modeling and a carefully defined sampling model.

9.9 Dependence Between Evidence Views

The six evidence views are conceptually separated but are not statistically independent. For example, photometric measurements can depend on image quality, image properties can correlate with observing conditions, and external matches can depend on source brightness and positional quality. For this reason, the analysis does not multiply evidence probabilities or assume that agreement among views automatically represents independent confirmation. The views are treated as complementary diagnostic information.

10 Reproducibility and Provenance

Reproducibility was treated as a scientific requirement rather than as an afterthought. Each major experimental stage produced machine-readable outputs, and the final analysis freeze recorded experiment provenance, source checksums, candidate results, and consistency checks.

The final source-level analysis contained 175 sources and 52 columns in the complete evidence representation. The canonical machine-learning feature matrix was reduced to 22 value features and two availability indicators, avoiding direct inclusion of validator outcomes and other identity or leakage variables.

The final freeze identified six validator-positive sources, two Tier 1 candidates, two Tier 2 candidates, and two control/sanity sources. No hard analysis failures were identified in the final freeze, while two documented warnings remained associated with known limitations.

The final candidate master was:

Group	Validator χ^2_ν	Phase VII χ^2_ν	LOO minimum	Tier	Status
1978	3.9392	4.1404	2.7985	TIER 1	Not confirmed
1974	3.0262	3.0473	2.1420	TIER 1	Not confirmed
233	2.1185	–	1.6531	TIER 2	Unresolved
1976	2.1159	–	1.8238	TIER 2	Unresolved
234	2.4932	–	–	CONTROL	Sanity control
1979	2.2789	–	–	CONTROL	Sanity control

A small numerical discrepancy was documented between the earlier Phase V candidate measurements and the later Phase VII direct recomputation. For group 1974, the W1 reduced chi-square changed from approximately 3.0262 to 3.0473. For group 1978, it changed from approximately 3.9392 to 4.1404.

These differences were not silently overwritten. They arose from differences in the later direct reconstruction and were retained in the provenance record. Importantly, both sources remained above the operational threshold and retained their qualitative investigation status.

The analysis freeze was therefore designated:

ANALYSIS-FREEZE-READY-WITH-DOCUMENTED-LIMITATIONS.

This status means that the current exploratory result is sufficiently documented to support independent review of the implemented analysis, while not implying that the scientific candidates themselves have been confirmed.

11 Future Work

The next stage of the project should prioritize independent validation rather than increasing model complexity.

First, the frozen pipeline should be applied to an independently selected field that was not used during candidate development. This would provide a more meaningful test of whether the same evidence patterns recur outside the development region.

Second, the external catalogue stage should be expanded and made robust against query failures. Gaia, SIMBAD, GCVS, and additional infrared or optical survey products could provide independent identities, classifications, and light curves.

Third, the highest-priority candidates should receive more complete image-level analysis, including the official artifact masks and detector-coordinate information. The aim should be to test specific instrumental hypotheses rather than simply inspect additional images.

Fourth, targeted follow-up observations would provide the strongest route toward confirmation. Independent observations at different times, wavelengths, instruments, or observing geometries could distinguish source-intrinsic behavior from NEOWISE-specific effects.

Fifth, if an independent validation population becomes available, the statistical framework could be extended to formal false-discovery control and calibrated candidate probabilities. Such extensions should follow, rather than precede, the establishment of independent labels and a frozen evaluation protocol.

Finally, the unsupervised learning stage could be expanded only where additional complexity answers a specific scientific question. More sophisticated models should not be introduced merely to increase model complexity or anomaly-score resolution.

12 Conclusion

This work developed and tested an evidence-first framework for investigating unusual astronomical sources in NEOWISE single-exposure data.

The central principle is that statistical unusualness is not equivalent to discovery. The pipeline therefore separates source-history construction, uncertainty-aware temporal screening, unsupervised novelty triage, observational/systematic checks, external-catalogue evidence, and astrophysical interpretation.

Applied to a small development field, the workflow reduced 8,652 retrieved detections to 7,275 working detections, 3,371 source-like groups, 175 sufficiently sampled source histories, and six operational validator positives. Subsequent multi-view analysis identified groups 1978 and 1974 as the highest-priority unresolved candidates. Both show elevated W1 temporal scatter that remains above the operational robustness threshold after individual observations are removed.

The evidence does not establish intrinsic astrophysical variability. W2 does not reproduce the same strength of behavior, nearby sources remain present, image-level measurements provide only source-associated evidence, detector-level effects were not fully tested, and external identities remain unresolved. The two candidates are therefore classified as high-priority follow-up targets rather than confirmed discoveries.

The study also demonstrates several practical failure modes of astronomical anomaly detection.

Brightness and signal-to-noise can dominate unsupervised novelty rankings; operational validator thresholds can strongly determine candidate selection; control populations can be mistaken for ground truth; missing external evidence can be incorrectly interpreted as negative evidence; and image-level differences can be mistaken for proof of intrinsic variability.

The resulting scientific position is deliberately conservative:

Unusual $\neq$ Novel $\neq$ Real $\neq$ Confirmed.

Within the present development analysis, groups 1978 and 1974 satisfy the criteria for high-priority unresolved investigation. Groups 233 and 1976 remain secondary candidates, while groups 234 and 1979 provide control-compatible sanity checks. No new astronomical class or physical phenomenon is established.

The principal contribution is therefore not a claim of discovery, but a reproducible investigation strategy in which anomaly detection is treated as the beginning of scientific inquiry rather than its conclusion.

13 Data and Code Availability

The public repository containing the research code, selected notebooks, methodology documentation, and canonical manuscript is available at:

<https://github.com/rahmandarji/evidence-first-astronomical-anomaly-detection>

The public research release is also archived through Zenodo and assigned the persistent DOI:

<https://doi.org/10.5281/zenodo.22841621>

The corresponding Zenodo record preserves the public research package associated with the **v1.0.1** release.

The underlying NEOWISE survey data and external catalogue data are not redistributed. Researchers should obtain those datasets from their respective public archives and reproduce only the data products required for the analysis.

The complete private research archive remains outside the public repository. It contains broader experimental history, intermediate artifacts, large data products, frozen evidence archives, and other development materials not required for the public code release.

The public GitHub repository represents the live research-code and documentation layer, while the Zenodo record provides a persistent archived version of the public release. Together, these resources define the public reproducibility layer of the project rather than a complete redistribution of the underlying survey data or private development archive.

Acknowledgements

References

- [1] Wright, E. L., Eisenhardt, P. R. M., Mainzer, A. K., et al. “The Wide-field Infrared Survey Explorer (WISE): Mission Description and Initial On-orbit Performance.” *The Astronomical Journal*, 140, 1868–1881, 2010. doi:10.1088/0004-6256/140/6/1868.
- [2] Mainzer, A., Bauer, J., Cutri, R. M., et al. “Initial Performance of the NEOWISE Reactivation Mission.” *The Astrophysical Journal*, 792, 30, 2014. doi:10.1088/0004-637X/792/1/30.
- [3] Paz, M., Kirkpatrick, J. D., Uttamchandani, R., Raen, T., & Cutri, R. M. “VarWISE: Infrared Variability via NEOWISE Single-exposure Photometry.” *The Astrophysical Journal Supplement Series*, 284, 41, 2026. doi:10.3847/1538-4365/ae562f.
- [4] NASA/IPAC Infrared Science Archive. “NEOWISE and VarWISE data and archive documentation.” California Institute of Technology, Pasadena, CA. <https://irsa.ipac.caltech.edu/>.
- [5] Lochner, M., & Bassett, B. A. “Astronomy: Personalised active anomaly detection in astronomical data.” *Astronomy and Computing*, 36, 100481, 2021. doi:10.1016/j.ascom.2021.100481.
- [6] Breunig, M. M., Kriegel, H.-P., Ng, R. T., & Sander, J. “LOF: Identifying density-based local outliers.” *ACM SIGMOD Record*, 29(2), 93–104, 2000. doi:10.1145/335191.335388.
- [7] Liu, F. T., Ting, K. M., & Zhou, Z.-H. “Isolation Forest.” *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining*, 413–422, 2008. doi:10.1109/ICDM.2008.17.
- [8] Rousseeuw, P. J., & Van Driessen, K. “A fast algorithm for the minimum covariance determinant estimator.” *Technometrics*, 41(3), 212–223, 1999. doi:10.1080/00401706.1999.10485670.
- [9] Gaia Collaboration. “Gaia Data Release 3: Summary of the astrometric, photometric, and survey properties.” *Astronomy & Astrophysics*, 674, A1, 2023. doi:10.1051/0004-6361/202243940.
- [10] Wenger, M., Ochsenbein, F., Egret, D., et al. “The SIMBAD astronomical database: The CDS reference database for astronomical objects.” *Astronomy and Astrophysics Supplement Series*, 143, 9–22, 2000. doi:10.1051/aas:2000332.